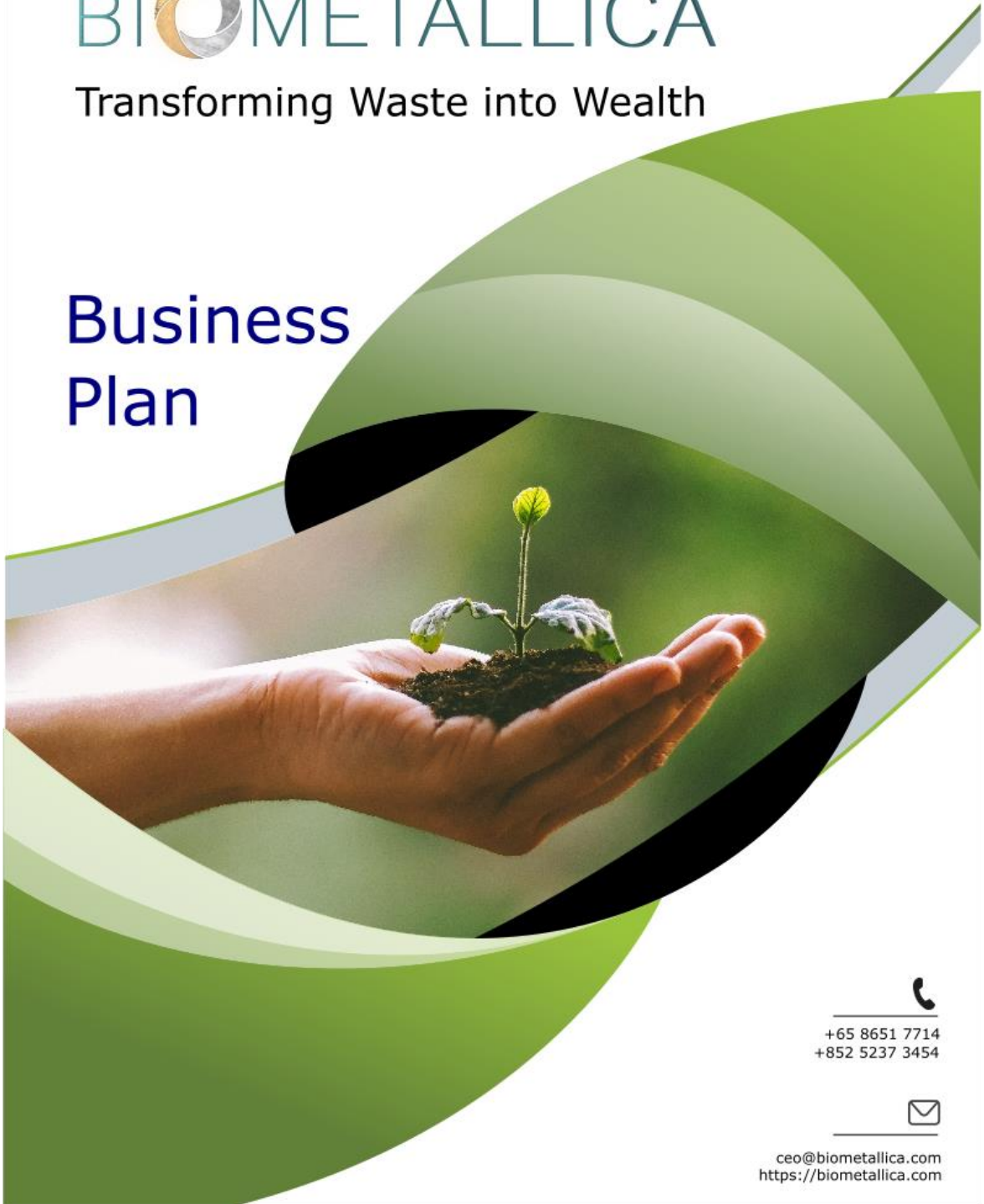




Transforming Waste into Wealth

Business Plan



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Executive Summary

BioMetallica emerges as a pioneering force in the recycling industry, focusing on the sustainable recovery of Precious Group Metals (PGMs) from spent auto-catalysts, electronic waste, and other industrial by-products and end-of-life products. Leveraging cutting-edge biometallurgical and electrolytic technologies, BioMetallica addresses the critical need for eco-friendly metal recovery solutions that significantly reduce environmental impact, offering a compelling alternative to traditional mining and recycling methods.

Vision and Mission

Our mission is to revolutionize the PGM recovery process, making it more sustainable, efficient, and cost-effective. BioMetallica aims to lead the transition towards a circular economy for precious metals, contributing to global efforts to reduce electronic waste and minimize carbon emissions. Our vision encompasses not only environmental stewardship but also the creation of value for our stakeholders through innovative technology and strategic market expansion.

Market Opportunity

The demand for PGMs, driven by their extensive applications in the automotive, electronics, and renewable energy sectors, continues to grow amid increasing environmental concerns and stringent regulations on recycling practices. BioMetallica targets this burgeoning demand, focusing on markets with significant e-waste generation and PGM use, including Southeast Asia, China, and Japan. Our market analysis identifies a serviceable obtainable market (SOM) valued at US\$7.8 billion, with BioMetallica aiming to capture a substantial share through our sustainable recycling solutions.

Products and Services

BioMetallica's core offering centers around our proprietary bioextraction process, capable of recovering high-purity PGMs while minimizing energy consumption, waste generation, and water usage. Our recycling-as-a-service (RaaS) cater to electronic manufacturers, automotive companies, pharmaceutical companies, implantable medical device manufacturers and catalysis research institutes, providing them with sustainably sourced high-purity metals essential for their operations. Additionally, our technology paves the way for collaborations with environmental agencies and waste collectors, further expanding our market reach.

Competitive Advantage

Our competitive edge lies in our biometallurgical technology, which is not only more environmentally friendly but also 6 times more energy-efficient and 3 times cheaper than conventional recycling methods. The unique expertise of our team, including visionary co-founders including Ing. Dr. Ansah-Antwi and Dr. Li, with backgrounds in decarbonizing and sustainable technologies and seasoned advisors like Mr. Rajeev Kumar and Mr. Sam Chow, with extensive experience in chemical and industrial engineering, positions BioMetallica as a leader in sustainable metal recovery.

Financial Projections

BioMetallica projects robust growth over the next five years, targeting a market share of 10-15% in our selected regions. With seed round funding requests of \$2-3 million to scale operations and enhance R&D efforts, we anticipate generating over \$50-100 million in annual revenue from recycling PGMs within Hong Kong alone at our maximum recycling capacity of 250 tons per year, expanding significantly as we penetrate further into our target markets.

BioMetallica stands at the forefront of a sustainable revolution in precious metal recovery. With a strong foundation in innovative technology, a clear understanding of market dynamics, and a team equipped with both technical expertise and commercial acumen, BioMetallica is poised to deliver substantial environmental and economic benefits, driving forward our mission of promoting a cleaner, greener future.

1. Business Startup Description

1.1 Mission Statement

Our mission is to provide innovative and eco-friendly precious metal recycling solutions that maximize the value of waste materials and minimize the environmental impact of metal production, while providing excellent customer service and creating a sustainable future for all.

1.2 Company Philosophy and Vision

BioMetallica's vision is to be a leader in sustainable precious metal recycling, creating a circular economy that benefits the environment and society. At BioMetallica, we are committed to driving sustainability and innovation in the precious metal recycling industry by providing efficient and eco-friendly open and closed loop recycling services. We aim to create value for our customers by delivering high-quality recycled precious metals and waste management solutions while reducing the environmental impact of the industry. Our team is dedicated to advancing technological solutions that address the evolving needs of our customers and contribute to a more sustainable future.

1.3 Industry Overview

The precious metal recycling industry, in which BioMetallica operates, is characterized by its growing nature, driven by the increasing demand for sustainable and environmentally friendly solutions. The industry has witnessed significant growth in recent years due to the rising awareness of the environmental impact of traditional mining methods and the need to conserve precious metal resources.

In the short-term, the industry is expected to continue growing at a compounded annual growth rate (CAGR) of about 10% as more companies and consumers embrace sustainable practices and seek alternatives to traditional mining and recycling. The long-term outlook remains positive as well, as the demand for precious metals continues to rise in various sectors such as electronics, automotive, biomedical and renewable energy. BioMetallica is well positioned to take advantage of these industry trends and changes. Our innovative and efficient biometallurgical processes offer a sustainable solution for precious metal recycling, aligning with the growing demand for environmentally friendly practices. By leveraging our unique technology, we can provide cost-effective and efficient recycling services to meet the increasing needs of customers in the industry.

Additionally, our closed-loop recycling process and ability to recover multiple metals simultaneously in a few steps set us apart from competitors, enabling us to capture a larger market share. As the industry moves towards more stringent environmental regulations and sustainability goals, BioMetallica's focus on minimizing environmental impact and reducing costs positions us as a leader in the market.

Furthermore, our commitment to continuous research and development allows us to stay at the forefront of industry advancements and adapt to evolving customer demands. By investing in technological innovation and process optimization using state-of-the-art internet of things (IoT) and artificial intelligence (AI), we ensure that our business remains competitive and capable of meeting the future needs of the industry.

Overall, BioMetallica is poised to thrive in the growing and evolving precious metal recycling industry by capitalizing on the demand for sustainable solutions, leveraging our unique technology, and remaining agile in response to industry trends and changes.

1.4 Legal Structure

BioMetallica HK Ltd is a legal entity structured as a private company limited by shares and incorporated under the laws of Hong Kong. It is operated as a wholly owned subsidiary of BioMetallica Pte Ltd, based in Singapore. BioMetallica is a spinoff of the National University of Singapore (NUS).

1.5 Business Startup Goals and Objectives

At BioMetallica, our goals and objectives are focused on driving innovation, sustainability, and growth in the precious metal recycling industry. We have outlined specific measurable goals with defined timeframes to guide our progress:

1. **Increase market share:** Achieve a 10-15% market share in the precious metal recycling industry within the next three-five years.
2. **Revenue growth:** Double our annual revenue within the next five years by expanding our customer base and increasing the volume of recycled precious metals to a maximum capacity of 250 tons per year per facility.
3. **Research and development:** Invest 20-30% of annual revenue into research and development activities to enhance our proprietary bioextraction technology and develop new sustainable solutions.

4. **Environmental impact:** Reduce year-on-year carbon emissions by 20% within the next five years through the implementation of energy-efficient processes and the adoption of renewable energy sources.
5. **Customer satisfaction:** Maintain a customer satisfaction rate of at least 90% based on regular surveys and feedback.

2. Products/Services

BioMetallica as sustainable manufacturing pioneer specializes in the following top three products and services:

1. **Biorecovery Technology:** Utilizing advanced genetic engineering, BioMetallica offers a groundbreaking biorecovery process, enhancing cyanide production in *Chromobacterium violaceum* by 100% compared to the wild-type. This technology efficiently recovers precious metals, such as platinum, palladium, and gold, from electronic waste and auto-catalysts.
2. **Surface Modification Techniques:** The company employs state-of-the-art surface modification techniques, including slurry electrolysis to produce high-quality nanoparticles and thin films. This nanofabrication is crucial for biomedical device coatings and applications in cancer drug manufacturing.
3. **Customized Bioreactors and Electrolytic Systems:** BioMetallica designs and manufactures modular bioreactors and electrolytic systems tailored for industrial-scale applications. These customizable systems are scalable, accommodating varying quantities of feedstock for efficient precious metal recovery. These offerings collectively position BioMetallica as a leading provider of sustainable and innovative solutions in the precious metal recovery industry.

2.1 Products/Services Description

At BioMetallica, we specialize in the recycling and recovery of precious metals using our patented bioextraction and electrolytic technologies. We offer a range of sustainable and cost-effective solutions for the recycling of precious metals such as gold, silver, platinum, rhodium and palladium.

Our manufacturing process involves several key steps:

1. **Collection:** We work closely with strategic partners and suppliers to collect end-of-life materials, such as electronic waste, spent auto-catalysts, and other sources rich in precious metals.

2. **Mechanical particle size reduction and ultrasonication pre-treatment:** The particles sizes are reduced to sub 10 microns in diameter for efficient pre-treatment using ultrasonication in a chemical bath to separate the precious metals from the non-precious metals for high efficiency bioleaching step.

3. **Bioextraction and electrolytic process:** Our proprietary bioextraction technology is utilized to extract and recover precious metals from the materials. This innovative process minimizes environmental impact and produces zero wastewater. The electrolytic process involves the use of a glycol-based, pH-neutral electrolyte that is 100% recyclable. Our innovative method operates at a low current density and is highly selective for different metals.

4. **Refining:** The extracted precious metals are further refined to ensure high purity and quality.

5. **Product Delivery:** We work with manufacturers and suppliers to deliver recovered precious metals to various industries, including electronics, battery, jewelry, medical devices and automotive.

To ensure a smooth and efficient product delivery, we have established strategic relationships with manufacturers, suppliers, and partners in the precious metal recycling industry and waste management supply and value chains. These collaborations enable us to access a steady supply of raw materials, optimize our manufacturing processes, and meet the specific requirements of our customers. By maintaining strong relationships and partnerships, we can ensure the timely delivery of our products/services and provide exceptional value to our clients.

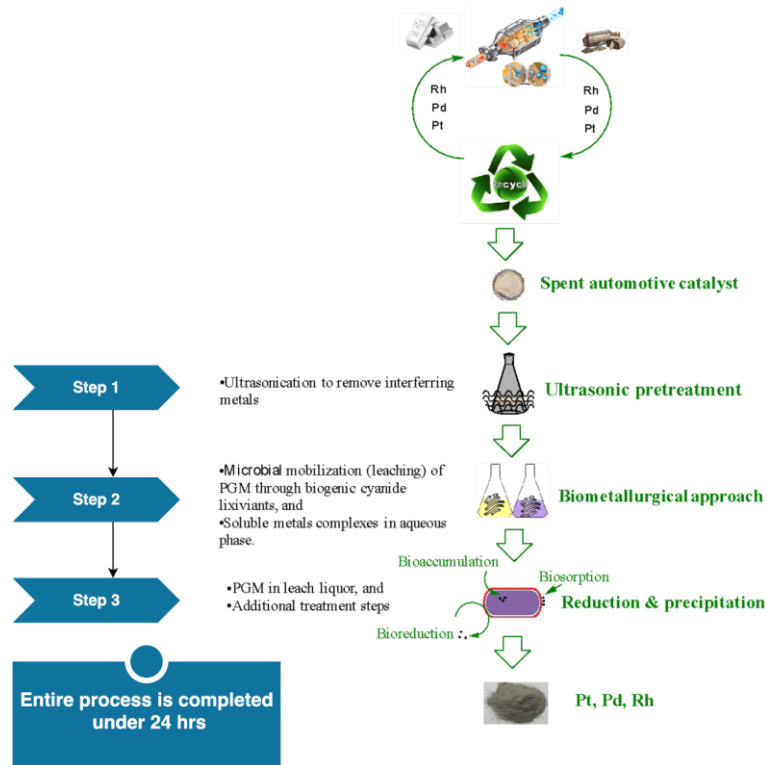
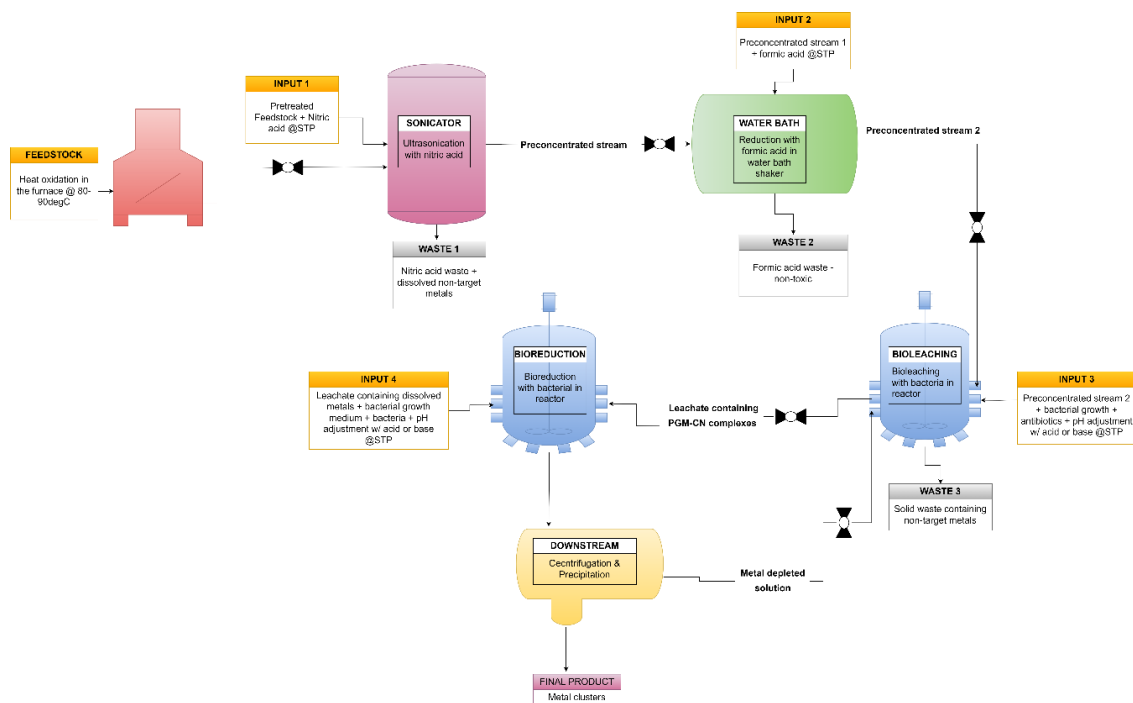


Illustration of the BioMetallica's proprietary bio-extraction process to recycle and recover PGM from spent autocatalyst.



Process Flow Diagram of BioMetallica's bioextraction process.

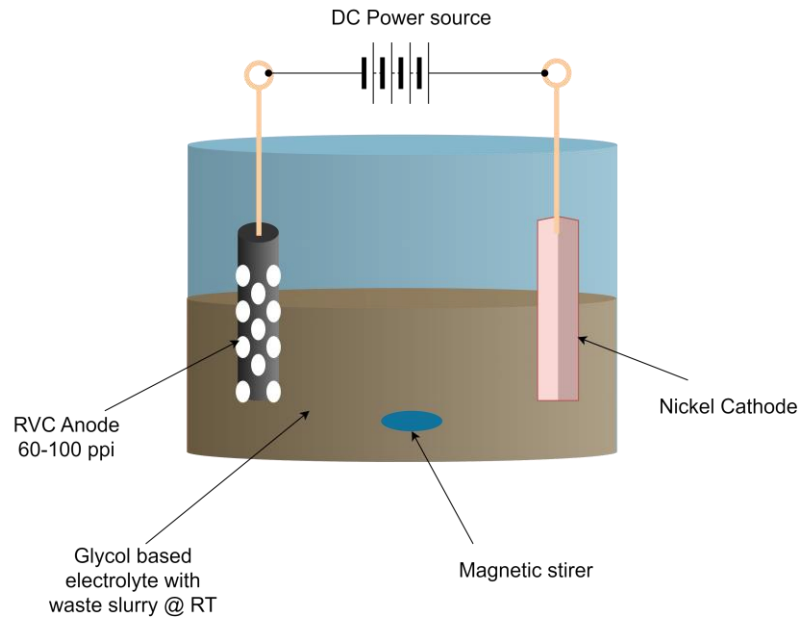


Illustration of BioMetallica's slurry electrolysis process.

2.2 Unique Features or Proprietary Aspects

BioMetallica offers unique value-added characteristics that set us apart and provide a competitive advantage in the precious metal recycling industry.

1. **Sustainable and Eco-friendly:** BioMetallica's bioextraction technology revolutionizes the recycling of precious metals, championing sustainability and environmental stewardship. Our process significantly lowers energy use, diminishes waste production, and eliminates wastewater, marking a substantial stride towards a cleaner, greener planet. Impressively, for every ton of waste processed, our technology prevents 44 tCO₂e of carbon emissions, setting a new standard in eco-friendly recycling practices compared to conventional methods.
2. **Cost-effective:** Our innovative processes and efficient extraction methods result in 3x cost savings compared to traditional recycling methods. By offering a more cost-effective solution for precious metal recycling, we provide our customers with competitive pricing and improved profitability.
3. **High energy efficiency:** BioMetallica's technology sets a new benchmark in the precious metal recovery industry with its remarkably low energy budget, operating on a fraction of the power required by traditional methods. This groundbreaking process achieves an impressive 6 times energy efficiency, drastically reducing environmental impact and operational costs, and positioning BioMetallica as a leader in sustainable innovation.

4. **Customization and Flexibility:** We understand that different customers have unique requirements. We offer customization options to tailor our services and products to meet specific needs. This flexibility allows us to address diverse customer demands and build long-term relationships.
5. **Expertise and Innovation:** With a team of experienced professionals and researchers, we continuously invest in research and development to enhance our bioextraction and electrolytic technologies and improve our processes. Our commitment to innovation ensures that we stay at the forefront of the industry, offering the latest advancements in precious metal recycling.

By combining these value-added characteristics, BioMetallica positions itself as a reliable and forward-thinking partner in the precious metal recycling sector. Our sustainable practices, cost-effectiveness, high-quality products, flexibility, and expertise give us a competitive edge, attracting customers who prioritize environmental responsibility, economic efficiency, and premium materials.

2.3 Problem Solved by Product/Service

Market Problems:

1. **Environmental Impact:** Traditional recycling methods for precious metals lead to considerable environmental pollution, marked by the emission of toxic gases and the creation of secondary liquid waste laden with harmful heavy metals.

- **Solution:** BioMetallica addresses these challenges with its sustainable and eco-friendly bioextraction technology, which drastically lowers energy use, curtails waste production, and completely eradicates wastewater output. Our approach underscores a commitment to responsible resource management and environmental preservation. Furthermore, our innovative electrolytic extraction method employs a 100% recyclable glycol-based electrolyte, in stark contrast to the strong acids utilized in conventional electrolysis processes, enhancing our sustainability credentials.

2. **Rising Costs:** The cost of mining and refining precious metals continues to escalate due to depleting ore reserves, increasing labor costs, and stricter environmental regulations.

- **Solution:** BioMetallica's cost-effective recycling solution offers a more economical alternative to traditional methods. By recovering precious metals from secondary sources like spent auto-catalysts and electronic waste, we reduce the reliance on costly mining operations, making our services financially advantageous for customers.

3. **Supply Chain Disruption:** The global supply of precious metals can be unstable due to geopolitical tensions, trade restrictions, and market fluctuations, leading to supply chain disruptions for industries reliant on these materials.

- **Solution:** BioMetallica provides a reliable and consistent supply of high-quality precious metals through our recycling processes. By offering a circular economic approach through urban mining, we reduce dependence on volatile supply chains, ensuring a steady and secure source of raw materials for our customers.

2.4 Production

Critical Factors in the Production of BioMetallica's Products/Services:

1. Capital Equipment:

- **Bioreactors and Electrolytic Systems:** Customized modular bioreactors and electrolytic systems are at the core of BioMetallica's operations, driving both the recovery of precious metals and the application of precious metal coatings on medical devices. Designed with scalability, efficiency, and adaptability in mind, these systems can handle varying quantities of feedstock, ensuring optimal performance across diverse operational scales.

2. Materials:

- **Feedstock (E-Waste and Auto-catalysts):** The availability and quality of electronic waste and auto-catalysts are critical material inputs. Securing a consistent supply through strategic partnerships with waste collectors is essential for uninterrupted operations.

- **Genetically Engineered Organisms:** The production of biogenic cyanide relies on the genetic engineering of *Chromobacterium violaceum*. The stability and performance of these genetically modified organisms are crucial factors.

3. Labor:

- **Genetic Engineers and Biochemists:** Skilled scientists and engineers are required to oversee the genetic engineering process and optimize biorecovery conditions.

- **Technicians for Surface Modification:** Expertise in operating and maintaining electrolytic systems and other surface modification equipment is essential.

- **Process Engineers:** To monitor and optimize the overall production process, ensuring efficiency and maximizing precious metal recovery rates.

4. Strategic Partnerships:

- **Waste Collectors and Precious metal waste suppliers:** Building and maintaining partnerships with waste collectors are crucial for a steady flow of electronic waste and auto-catalysts, which are vital for our operations. BioMetallica has effectively forged relationships with leading suppliers, such as

iZ Precious in Italy and BR Metals Ltd in Singapore, guaranteeing a dependable supply of precious metal waste for our recycling processes.

5. Certification and Compliance:

- **Certification Organizations:** Collaboration with entities like Gold Standard for certification of sustainable and carbon reduction technologies is integral for market credibility.

2.5 Product/Service Pricing

BioMetallica adopts a strategic pricing model based on the value proposition of its innovative products and services. The pricing structure is designed to reflect the unique features, sustainability, and technological advancements offered by BioMetallica in the competitive landscape of precious metal recovery for biomedical applications. The pricing, subscription, or fee structure is outlined as follows:

1. **Fee for Biorecovery Services:** The company charges a fee for its biorecovery services, encompassing the genetic engineering of *Chromobacterium violaceum*, the bioremediation process, and the recovery of precious metals from electronic waste and auto-catalysts.
2. **Equipment and Technology Licensing:** BioMetallica offers licensing agreements for its proprietary technologies, including biorecovery processes, surface modification techniques, and customized bioreactors. Clients may pay licensing fees for the use of these technologies in their own operations.
3. **Metal Trading Value-Based Pricing:** The pricing strategy aligns with the unique value proposition of BioMetallica's products and services, taking into account the eco-friendly nature of the processes, the high purity of recovered metals, and the potential cost savings for clients in comparison to traditional mining and processing methods.

2.6 Manufacturability and Scalability:

Designed for scalability, our technology features expandable modular bioreactors and established partnerships with engineering and manufacturing firms, ensuring readiness to meet the surging demand for green metal recovery solutions. We aim to establish facilities capable of processing up to 250 tons/year of feedstock, with strategic CapEx investments of \$103 million and projected revenue of \$220 million. It will take about five and a half years to reach the initial capacity of 250 tons per year. From the sixth year onward, we plan to double this capacity each year.

2.7 Novelty of Proposed Approach & Product In Relation to Competitors

BioMetallica's methodology for recovering precious metals from electronic waste and spent auto-catalysts fundamentally diverges from conventional techniques utilized by industry incumbents. Traditional methods such as pyrometallurgy and hydrometallurgy depend on substantial energy inputs and the application of aggressive chemical solvents, respectively. These approaches not only have a high energy footprint but also pose considerable environmental risks through toxic emissions and the generation of hazardous wastes. Entities like Tanaka Holdings and Umicore have largely adopted these conventional practices, which, despite their effectiveness, are increasingly scrutinized for their environmental and energy implications. Emerging entities, including PH7, Mint Innovation, and Phoenix Tailings, present eco-friendlier alternatives but are often restricted by the specificity of feedstock and metal diversity they can process. Specifically, Mint Innovation and Phoenix Tailings leverage microorganisms for selective extraction of gold and rare earth elements, respectively, limiting their feedstock versatility. PH7, targeting platinum group metals similar to BioMetallica, employs solvometallurgy, a process that, while reducing water usage, generates considerable volumes of organic waste.

Bioleaching Innovation:

BioMetallica's bioleaching technology represents a paradigm shift, utilizing genetically engineered microorganisms for the production of biogenic cyanide. This innovative approach allows for the formation of stable metal-cyanide complexes at ambient temperatures, facilitating the extraction of a broader spectrum of precious metals. This method reduces energy requirements and diminishes environmental hazards, positioning BioMetallica at the forefront of sustainable metal recovery technologies. Notably, our process accommodates pulp densities exceeding 4%, surpassing the operational limits of traditional non-hydrometallurgy and pyrometallurgy methods.

Overcoming Historical Limitations:

Previous ventures into bioleaching and alternative eco-friendly methodologies have frequently encountered scalability and efficiency obstacles. BioMetallica's use of engineered microorganisms surmounts these challenges, optimizing bioleaching efficiency and metal

recovery at elevated pulp densities. This advancement ensures environmental sustainability and commercial scalability.

Comparative Advantages:

- **Energy Efficiency:** Our processes exhibit a sixfold increase in energy efficiency compared to conventional recovery methods, addressing critical concerns around excessive energy use.
- **Environmental Impact:** Zero wastewater production and reduced chemical reliance establish a new benchmark in minimizing the ecological footprint of metal recovery operations.
- **Scalability and Economic Feasibility:** Demonstrated scalability and a threefold reduction in operational costs position BioMetallica as a viable and cost-effective alternative for industry adoption.

2.8 Impact of BioMetallica's solution and business on the Society

HK Tech 300 Angel fund support is crucial for BioMetallica's aim to transform the recycling landscape in Hong Kong and globally. Our project on sustainable precious metal recovery from e-waste aligns with Hong Kong's zero-waste ambitions and environmental goals. Support from HK Tech 300 Angel Fund and other government funding support would accelerate the development of a groundbreaking solution to the pressing issue of electronic waste.

As a technological hub, Hong Kong faces significant e-waste challenges, which BioMetallica addresses by converting waste into valuable resources. This approach supports Hong Kong's Green Plan 2030 by minimizing landfill use, reducing greenhouse emissions, and lessening reliance on mining, thus bolstering Singapore's leadership in green technology.

Our technology promises local economic growth through green job creation and positions Hong Kong as an innovator in sustainable recycling on the world stage. Beyond environmental benefits, BioMetallica enhances public health and offers scalable solutions to waste management across various sectors.

HK Tech 300 backing would mark a strategic investment in Hong Kong's sustainable development, showcasing a commitment to pioneering a greener, more prosperous future through innovative e-waste management.

3. The Market

This section gives important details on the business's industry. It includes the competitive landscape, the target market of the business, and how to market to the customers. BioMetallica's technological breakthroughs offer a compelling, sustainable alternative to existing metal recovery practices. By outperforming traditional methods in key performance metrics such as energy efficiency, environmental preservation, and recovery efficacy, BioMetallica is poised to redefine the precious metal recovery sector. Our technology not only propels the recycling industry forward but also aligns with global initiatives for a sustainable, circular economy, making a significant contribution to the advancement of green technological solutions.

3.1 Industry Analysis

BioMetallica operates within the broader industry of precious metal recovery and sustainable technologies, with a specific focus on catering to the biomedical sector. The industry landscape is characterized by several dynamics, challenges, and opportunities that potential stakeholders, investors, and loan officers should understand:

Industry Dynamics:

- 1. Growing Demand for Precious Metals:** The biomedical sector has a rising demand for precious metals like platinum, palladium, and gold for use in medical devices, cancer treatment, and drug delivery systems. BioMetallica positions itself to capitalize on this growing market.
- 2. Sustainability and Environmental Concerns:** There is an increasing emphasis on sustainable practices and eco-friendly technologies in the precious metal industry. BioMetallica addresses environmental concerns by offering a biorecovery process that reduces the environmental impact associated with traditional mining.
- 3. Technological Advancements:** The industry is marked by continuous technological advancements in biorecovery and surface modification techniques. BioMetallica leverages cutting-edge

technologies, such as genetic engineering, slurry electrolysis, and pulsed laser deposition, to stay competitive.

4. Regulatory Compliance Challenges: Meeting regulatory standards, especially in industries like biomedical applications, poses a challenge. BioMetallica navigates these challenges by actively participating in certification programs and supporting clients in FDA approval processes.

5. Market Competition: The industry is competitive, with various players offering different approaches to precious metal recovery. BioMetallica distinguishes itself through its innovative biorecovery process, surface modification techniques, and commitment to sustainability.

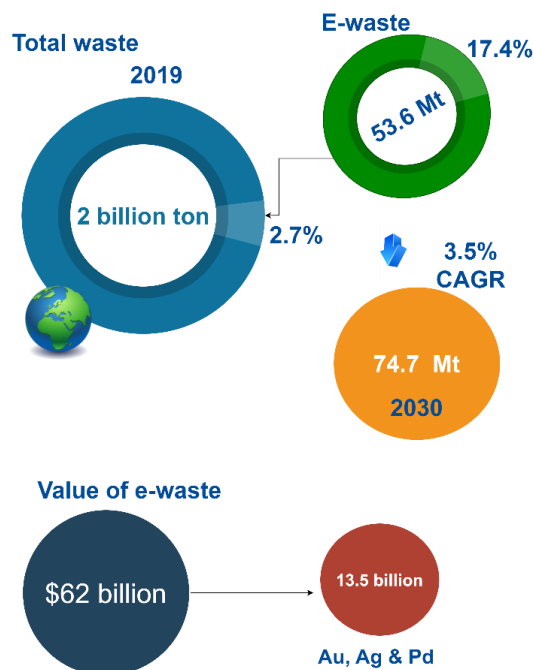
3.2 Market Analysis

Focus on e-waste:

- Annual e-waste generation: The world generated 53.6 Mt of e-waste in 2019. That's an average of 7.3 kg per capita. The amount of e-waste generated is expected to grow at about 3.5% per year and will reach 74.7 Mt by 2030. (2)
- E-waste recycling rate: In 2019 only 17.4% of the e-waste was collected and recycled. At the moment, the recycling activities are not keeping pace with the global growth of e-waste. (2)
- Value of the e-waste: The value of raw materials in the e-waste generated in 2019 was roughly USD 57 billion with only USD 10 billion recovered in an environmentally accepted way. Based on the Global e-waste monitor, gold, silver and palladium alone represent about USD 13.5 billion in value. (2)
- E-waste compared to the total waste: The world generates about 2 billion tonnes of waste annually of which 53.6 Mt in 2019 was e-waste. This means that e-waste represents about 2.7% of the total waste produced globally. (2)
- E-waste is not the best waste for a landfill: Plastic in the e-waste can take up to 1 million years to decompose, while aluminum and other metals can take anywhere between 50 and

500 years to break down. Some components in the e-waste will pollute the environment rather than biodegrade. (3)

- Mobile phones are a significant source of e-waste: Based on the Environmental Protection Agency report from 2014, in the US alone about 152 million mobile phones are discarded every year. (14)
- E-waste manufacturing is very resource-hungry: It takes at least 240kg of fossil fuels, 22kg of chemicals, and 1.5 tonnes of water to manufacture a single computer and its screen according to a UN study. This is more than the weight of a car. (15)
- E-waste represents a significant opportunity for circular economies: In 2019 seven UN entities released a report stating that annual e-waste production is worth over USD 62 billion annually. With only about 17% of e-waste recycled today, precious metals represent a significant value currently locked in the accumulated millions of tonnes of electronic waste. (16)



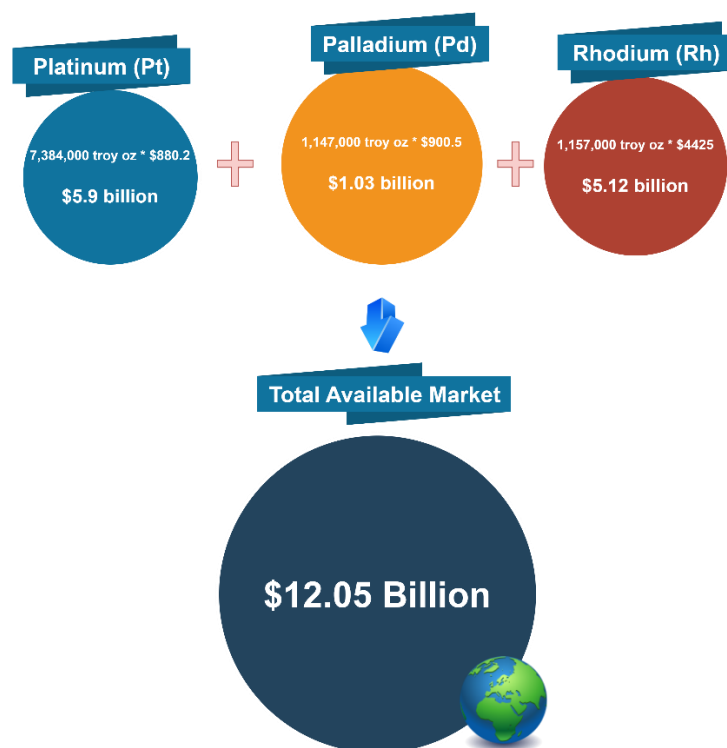
Focus on spent auto-catalyst:

BioMetallica's operations and technological advancements are presently centered around extracting platinum (Pt), palladium (Pd), and rhodium (Rh) from spent auto-catalysts, key components of the platinum group metals (PGMs). Utilizing a bottom-up approach, we've assessed the global demand for these essential PGMs, crucial for auto-catalytic applications. The average annual demand is quantified in millions of troy ounces (t oz) as follows: platinum at 7.4, palladium at 1.15, rhodium at 1.16, alongside gold at 152.4 and silver at 953.3.

Like all commodities, PGMs are subject to fluctuating market values, trading at spot prices. The latest 7-day moving average reveals spot prices per troy ounce of \$880.2 for platinum, \$900.5 for palladium, \$4425 for rhodium, \$2034.4 for gold, and \$440 for silver. This pricing translates to an aggregated market value of approximately \$12.05 billion, emphasizing the substantial economic impact and opportunity within the auto-catalytic sector for PGM recovery..^{1,2}

¹ Data Source : Johnson Matthey PGM Market Report, McKinsey Report and Statista reports on PGM market

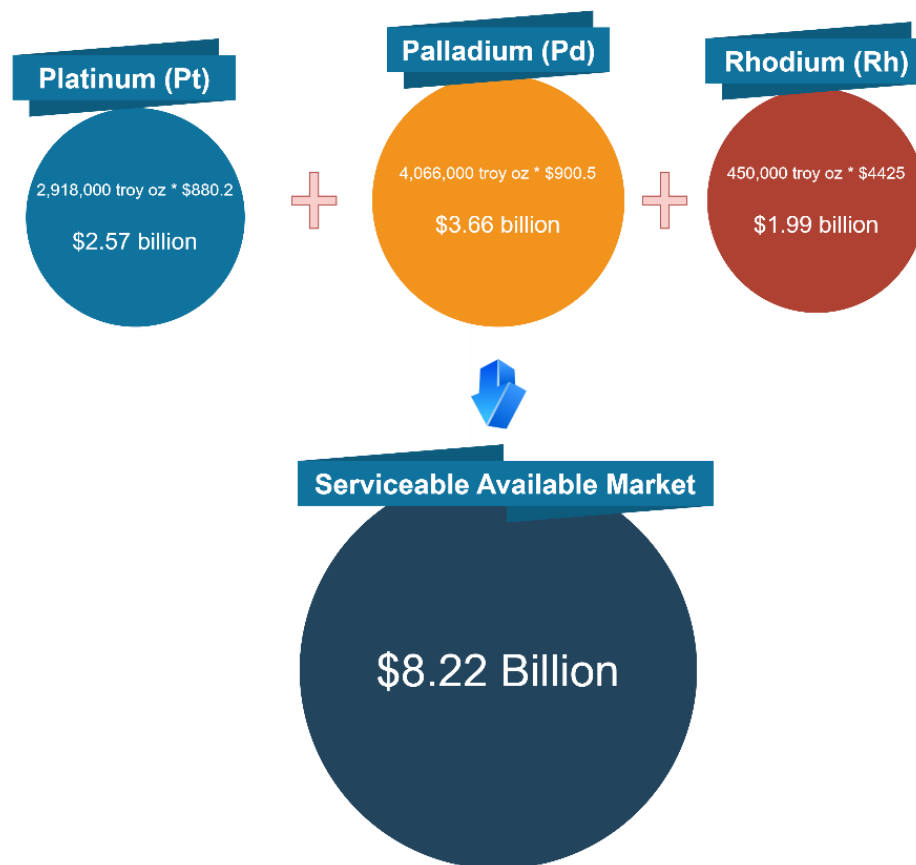
². [Daily Metal Price: Gold Price \(USD / Troy Ounce\) for the Last Month](#)



In alignment with BioMetallica's strategic direction for growth, we are focusing on expanding our footprint into the dynamic markets of Southeast Asia, China, and Japan. This expansion strategy is underpinned by a meticulous analysis of our Serviceable Available Market (SAM) and Serviceable Obtainable Market (SOM), utilizing both top-down and bottom-up approaches for precise estimation. Key to our market estimation are two principal sources of raw materials for Platinum Group Metals (PGMs) recycling: spent auto-catalysts and electronic waste (e-waste), pivotal in determining the breadth of SAM and SOM in these regions.

The market size, for both SAM and SOM, was derived by assessing the demand for PGMs — specifically Platinum (Pt), Palladium (Pd), and Rhodium (Rh) from spent auto-catalyst and Au and Ag from e-waste — alongside the annual production of relevant waste materials. Based on industry data, the demand within the SAM for Pt, Pd, and Rh is estimated to be 26,416 kg, 29,063 kg, and 8,445 kg, respectively. Leveraging the 7-day moving average spot prices for these metals, we've valued the SAM at an impressive US\$8.22 billion. This valuation not only signifies the substantial

market potential awaiting BioMetallica's innovative recycling technologies in these targeted regions but also reinforces our commitment to capturing and transforming waste into valuable resources, driving both economic and environmental benefits on a global scale.



BioMetallica, leveraging its unique biometallurgical recycling approach, is positioned to serve 10% of this target market. This approach not only stands out for its environmental and energy efficiency but also for its scalability at a lower capital expenditure compared to conventional methods. Specifically, in Singapore, with over 1 million vehicles equipped with three-way catalytic converters and robust e-waste collection infrastructure, the market potential is substantial. Detailed assessments, including customer interviews with e-waste collection firms, estimate Singapore and Hong Kong's market potential for BioMetallica at about US\$500 million annually. With 3 to 5 active competitors in the Singapore and Hong Kong recycling sector, BioMetallica is poised to capture over US\$100

million in annual revenue from recycling PGMs, thanks to our innovative, green, and energy-efficient technology. This strategic market positioning and technological advantage underscore BioMetallica's potential to make a significant impact in the PGM recycling industry, driving sustainable practices while capturing substantial market share.

In today's era, the potential and necessity for Platinum Group Metals (PGM) recycling have reached unprecedented heights. Platinum, being approximately 30 times rarer than gold, stands as a pivotal raw material, especially for hydrogen fuel cell energy generation—a cornerstone technology in the global pursuit of net-zero emissions. This unique positioning underscores the critical importance of PGM recycling, not only for conserving scarce resources but also for supporting sustainable energy solutions vital for our planet's future.

3.3 Competitor Analysis

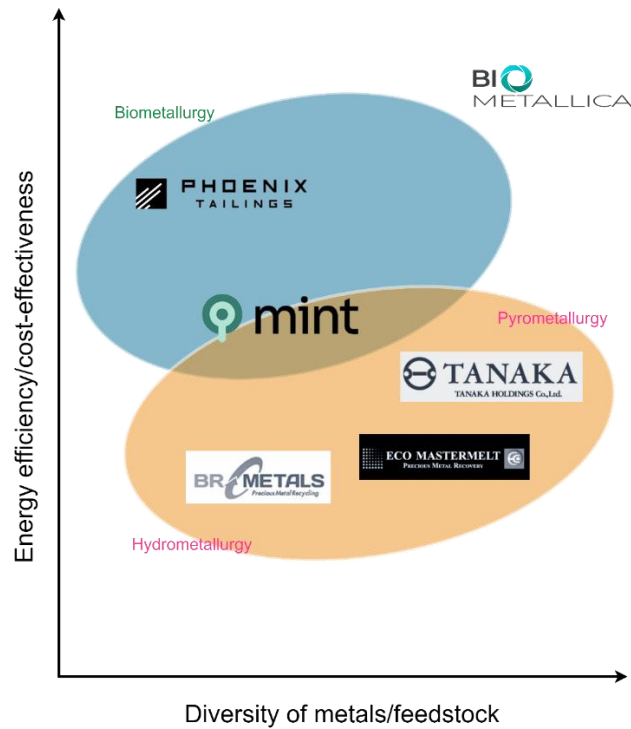
Mining and recycling serve as complementary sources in meeting the escalating demand for Platinum Group Metals (PGMs), crucial both as industrial raw materials and for investment purposes. Virgin PGM mining from natural ores is notably detrimental to the environment, generating significant greenhouse gas (GHG) emissions and demanding substantial energy. In contrast, traditional recycling methods, though more environmentally friendly than primary mining, produce less pollution and consume less water and energy. Yet, as the demand for these critical minerals surges to support the widespread applications of precious metals, it's imperative to overcome the limitations of current recycling processes to ensure a sustainable future for our planet.

Most existing recycling companies, including Tanaka Holdings, BR Metals, Eco-Mastermelt, and Heraeus, primarily utilize pyrometallurgy, hydrometallurgy, or a combination of both. A select few emerging companies, like Phoenix Tailings and Mint Innovation, partially incorporate biometallurgy into their proprietary recycling processes. A comparison of some key performance metrics among

the major players in the PGM recycling industry, highlighting BioMetallica’s contributions, is provided below:

	Yield	Scalability	Sustainability	Energy Efficiency	Cost Effectiveness
	High	High	Very High	Very High	Very High
	Moderate	Low	Very High	Very High	Very High
	Moderate	Low	Moderate	Low	Low
	High	Moderate	Low	Low	Low
	High	Moderate	Low	Low	Low
	Low	Moderate	Moderate	Low	Moderate

A cartesian plot of the cost effectiveness and energy efficiency of the competitive landscape for in BioMetallica’s launch market is shown graphically below



Analysis Summary:

- **Cost-Effectiveness:** BioMetallica and Phoenix Tailings are noted for their high cost-effectiveness, likely due to innovative, low-cost technologies. Tanaka Holdings scores lower, possibly due to more traditional, expensive processes.
- **Diversity of Feedstock:** BioMetallica and BR Metals stand out for accepting a wide range of feedstock, enhancing their flexibility and market reach. Eco Master Melt, with a lower score, may focus on more specific types of waste.
- **Recycled Metal:** Most companies offer a high diversity of recycled metals, with Mint Innovation slightly lower, suggesting a more focused or limited metal recovery capability.
- **Sustainability:** BioMetallica and Mint Innovation excel in sustainability, indicative of processes that minimize environmental impact, with Eco Master Melt and Phoenix Tailings also scoring high. Tanaka Holdings and BR Metals, while still committed, might use more conventional methods affecting their sustainability score.
- **Scalability:** BioMetallica's process is highly scalable, similar to BR Metals and Tanaka Holdings, suggesting a strong potential for expansion. Phoenix Tailings and Eco Master Melt might face more challenges in scaling up.

- **Reliability:** Tanaka Holdings is rated very high due to established, proven technologies.

BioMetallica also scores high, suggesting confidence in their process's consistency and effectiveness.

- **Yield of Processes:** High yields for BioMetallica, Phoenix Tailings, and Mint Innovation indicate efficient recovery processes. Eco Master Melt and BR Metals, with moderate scores, may face challenges in maximizing recovery.

This competitive analysis illustrates BioMetallica's strong position in the market, particularly in sustainability, cost-effectiveness, and the diversity of feedstock and recycled metals, highlighting its potential for growth and market penetration.

BioMetallica's biotechnological approach to precious metal recovery presents a compelling cost-effective alternative to traditional methods, driven by several key factors:

1. **Reduced Capital Expenditure:** Unlike conventional pyrometallurgy and hydrometallurgy techniques, which necessitate significant investment in specialized furnaces, advanced safety equipment, and chemical scrubbers and baths, BioMetallica's process requires less complex and expensive equipment and reagents. This fundamental difference in the technology setup translates to substantial savings in initial capital outlay, making the adoption and operation of BioMetallica's method more financially accessible.
2. **Lower Energy Costs:** The operation of BioMetallica's recovery process at lower temperatures is a critical factor in its cost-effectiveness. Traditional metal recovery methods often involve high-temperature processes that consume a significant amount of energy, leading to higher operational costs. BioMetallica's method, by contrast, reduces energy consumption considerably, offering a more sustainable and less costly alternative.
3. **Decreased Downstream Processing Costs:** The environmental and financial costs associated with wastewater treatment and the handling of toxic residues are notable drawbacks of traditional precious metal recovery methods. BioMetallica's process minimizes the production of toxic waste and requires less intensive downstream processing of wastewater streams. This not only leads to lower operational costs but also aligns with environmental compliance and sustainability goals, potentially reducing fees related to waste disposal and environmental impact.

3.4 Key Innovations

- **Genetically Engineered Microorganisms:** The employment of *Chromobacterium violaceum* strains, modified to augment cyanide production, boosts bioleaching effectiveness.
- **Optimization of Process Parameters:** Precise management of environmental conditions — temperature, dissolved oxygen levels, particle size, and pH — facilitates bioleaching efficiencies exceeding 90% at a solid-liquid mass ratio around 4%.
- **Advanced Pre-treatment Method:** This innovation effectively eliminates over 90% of base metals from the feedstock, mitigating interference in subsequent bioleaching stages.
- **Intelligent Process Monitoring:** Implementation of IoT (Internet of Things) technologies enables real-time tracking and optimization of the bioleaching operation, ensuring maximal metal recovery as seen in Figure 4.

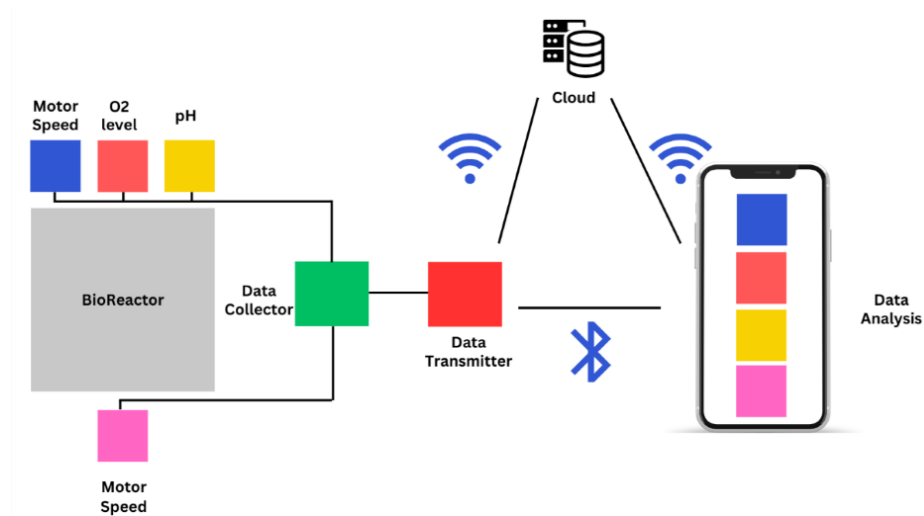


Figure 4 Illustration of data collection and analytics functionality for process optimization.

Significance of Innovations

These technological advancements mark a considerable progression from existing metal recovery techniques, introducing a more eco-friendly and economically viable alternative.

The innovation in microbial engineering to increase bioleaching efficiency is transformative, potentially diminishing dependence on primary metal extraction methods.

4. Marketing & Sales

4.1 Marketing Plan Overview for BioMetallica

In this chapter, we delve into the intricacies of BioMetallica's marketing plan, laying out a strategic roadmap designed to position us as frontrunners in the sustainable recovery of precious metals. Our plan is meticulously crafted around identifying and engaging with our target customer base, pinpointing precise market segments, and outlining the multifaceted strategies we will employ to achieve our objectives.

4.2. Target Customer Profile

BioMetallica's ideal customers are entities that recognize the value of sustainability and innovation in the recycling of precious metals. These include:

- **Precious metal waste collectors and recyclers:** Precious metal waste collectors and recyclers specialize in the aggregation, processing, and recovery of valuable metals from sources like spent auto-catalysts, electronic waste, and industrial by-products. These entities are pivotal in the supply chain of sustainable metal recovery, turning potential waste into valuable resources.
- **Electronics Manufacturers and Catalysis Research Institutes:** This segment includes electronics manufacturers who require high-purity metals for components such as semiconductors, conductors, and capacitors, where quality directly impacts product performance and reliability. Catalysis research institutes, on the other hand, focus on advancing catalytic technologies for various industrial applications, including energy conversion and chemical synthesis, where high-purity metals serve as essential catalysts.
- **Environmental Agencies:** Environmental agencies, encompassing both government and non-government organizations, are dedicated to promoting and enforcing sustainable industrial practices. Their scope of interest includes regulating emissions, managing waste, conserving resources, and encouraging the adoption of green technologies across various sectors.
- **Medical device coating and nanoparticles for anti-cancer drugs:** This segment encompasses companies and research institutions engaged in the development and manufacturing of medical devices and anti-cancer drugs. Medical device manufacturers utilize precious metals for their antimicrobial properties and biocompatibility, enhancing device safety and efficacy. Similarly, the pharmaceutical industry employs precious metal nanoparticles in the manufacturing of anti-cancer drugs, leveraging their unique properties to target and treat cancer cells more effectively.

4.3 Market Segmentation Strategy

BioMetallica's strategy within the precious metal recovery industry can be tailored to address the unique challenges and opportunities in each segment identified above. Let's break down a strategic approach for each key area:

1. Demand for Precious Metals in Biomedical Applications

- **Innovation in Recovery Techniques:** BioMetallica can focus on developing and refining biorecovery methods that are not only more efficient but also yield higher purity metals, which are critical for biomedical applications.
- **Partnerships with Biomedical Companies:** Establishing partnerships with companies involved in medical device manufacturing and cancer treatment could ensure a steady demand for recovered precious metals. Such collaborations can also lead to joint research and development efforts.
- **Certification and Compliance:** Obtaining certifications related to quality management and compliance with biomedical standards can enhance BioMetallica's reputation and reliability as a supplier of high-quality precious metals.

2. Electronic Waste Recycling and Auto-catalyst Recovery

- **Sustainable Recovery Processes:** Leveraging and promoting BioMetallica's eco-friendly recovery processes can appeal to companies and regulators looking to support sustainable practices in waste management and recycling.
- **Expansion of Collection Networks:** Developing a comprehensive network for collecting electronic waste and spent auto-catalysts can increase the input materials for precious metal recovery, ensuring a larger supply of metals.
- **Technological Investment:** Investing in technology to improve the efficiency of metal recovery from e-waste and auto-catalysts can reduce costs and environmental impact, positioning BioMetallica as a leader in sustainable recovery methods.

3. Sustainability and Eco-Friendly Recovery Methods

- **Sustainability Reporting and Communication:** Transparently reporting on BioMetallica's environmental impact and sustainability efforts can build trust with customers and regulators, highlighting the company's commitment to eco-friendly practices.
- **Certifications and Eco-Labels:** Pursuing sustainability certifications including Sustain Cert by Gold Standard and eco-labels can differentiate BioMetallica's products in the market, appealing to environmentally conscious consumers and businesses.

- **R&D for Green Technologies:** Continuous investment in research and development to improve the sustainability and efficiency of metal recovery processes can keep BioMetallica at the forefront of green technology in the industry.

4. Technological Advancements

- **Continuous Innovation:** Investing in ongoing research and development to refine existing technologies and explore new methods for precious metal recovery is crucial for maintaining a competitive edge.
- **Intellectual Property Protection:** Protecting innovations through patents and trademarks can secure BioMetallica's market position and allow for licensing opportunities.
- **Collaboration with Tech Companies:** Partnering with technology companies, especially those specializing in biotechnology and materials science, can accelerate the development of new recovery methods and applications for precious metals.

By adopting these strategic approaches tailored to each segment, BioMetallica can navigate the complexities of the precious metal recovery industry, especially within the electronics, battery manufacturing, precious metal recycling and biomedical field, and achieve sustainable growth and market leadership.

4.4. Strategies for Market Penetration

To penetrate these segments and engage our target customers, BioMetallica will implement a comprehensive approach:

- **Digital Marketing:** Utilizing SEO, targeted online advertising, and social media campaigns to raise awareness and educate potential clients about the benefits of our recycling solutions.
- **Direct Sales and Networking:** Participating in industry conferences, workshops, and exhibitions to build relationships with key stakeholders and decision-makers.
- **Strategic Partnerships:** Collaborating with e-waste collectors, auto-catalyst suppliers, and manufacturers to ensure a steady supply of raw materials and to expand our reach within the industry.
- **Content Marketing:** Developing informative content that highlights the importance of sustainable metal recovery and showcases BioMetallica's technological innovations and success stories.

4.5 IP Strategy

BioMetallica's intellectual property (IP) strategy is meticulously designed to safeguard our proprietary technologies and reinforce our competitive position in the sustainable precious

metal recovery sector. With an acute awareness of the pivotal role IP plays in our enduring success, our approach to IP management is comprehensive, encompassing the rigorous protection, strategic expansion, effective management, and judicious deployment of our IP assets. Currently, we hold exclusive licenses for two critical patents from the National University of Singapore, encapsulating genetic modification protocols for *Chromobacterium Violaceum* and advanced bioleaching processes, including response surface methodologies for process optimization.

Patent Portfolio Development and Protection: At the forefront of our IP strategy is the diligent protection and systematic expansion of our patent portfolio. This effort involves securing our foundational bioleaching patents while actively seeking to patent new discoveries and enhancements. We rigorously evaluate R&D outcomes for patentable innovations, ensuring comprehensive protection across key markets and jurisdictions that are strategically significant for our commercial objectives.

IP Acquisition and Portfolio Management: Collaborating with specialized IP legal experts from the National University of Singapore and City University of Hong Kong, we aim to acquire robust, defensible patents that afford broad coverage for our innovations. Our proactive patent management approach involves regular portfolio assessments to ensure alignment with our strategic business objectives, facilitating informed decisions on patent maintenance, expansion, and potential portfolio optimizations.

Strategic Licensing for Revenue and Partnerships: Our IP strategy incorporates an aggressive licensing model, aiming to monetize our technologies through selective partnerships in ancillary markets. These licensing endeavors are structured to uphold our environmental and quality standards while generating supplementary income, ensuring control over the primary applications of our technology.

IP Defense and Enforcement: Committed to defending our technological advancements and market presence, BioMetallica adopts a robust stance against IP infringement. This includes vigilant market surveillance for potential violations and assertive legal action as necessary.

We proactively communicate our IP rights and readiness to enforce them, deterring infringement and underscoring our commitment to protecting our innovations.

Alignment of IP Strategy with Business Objectives: Central to our approach is the seamless integration of our IP strategy with our overarching business goals. This dynamic alignment is continually evaluated and adapted in response to evolving business strategies, technological breakthroughs, and shifts in the competitive and regulatory landscapes.

In essence, BioMetallica's IP strategy is a cornerstone of our business model, crafted to secure and leverage our pioneering technologies in sustainable metal recovery. This strategic framework not only safeguards our innovations but also fuels growth, fosters strategic collaborations, and underpins our long-term leadership in the industry.

4.6. 5-year Milestones and Operational Plan

Our objective is to secure a 10-15% share of the precious metal recycling market in our targeted regions within the first five years. To achieve this, our marketing plan includes short-term goals focusing on brand visibility and network expansion, and long-term goals aimed at solidifying our market presence and expanding our customer base. Each phase of the plan is accompanied by specific milestones and performance metrics to monitor progress and adjust strategies as needed. Below is our 5-year operational road map and milestones:

Year 1: 2024 - Foundation and Design

Q3-Q4:

- Team Hiring: Recruit key project and technical leads including a bioprocess engineer, microbiologist, electrochemical engineer, and separation scientist/engineer.
- Fundraising: Secure seed funding including HKSTP Incubation programme and HK Tech 300 Angel Fund for initial design work, team salaries, and lab setup for early-stage research.
- Business Development: Initiate relationships with potential industry partners and suppliers.
- Research & Development:
 - Strengthen the core bioextraction processes by enhancing the capabilities of the existing 10-litre bioreactor system. Focus on genetic modification of microorganisms to increase metal yield.

- Initiate the development of a slurry electrolysis prototype that integrates seamlessly with the bioextraction process for enhanced metal recovery.
 - Enhance the prototype based on initial test results, focusing on energy efficiency and selectivity in metal recovery.
- IP Strategy: Secure intellectual property rights for unique processes and technologies developed.

Year 2: 2025 - Process Optimization and Integration

Q1-Q2:

- Fundraising: Conduct a pre-Series A funding round to finance the optimization and development of the bioleaching and electrolysis processes.
- Team Hiring: Expand the R&D team by adding specialists in electrolysis, environmental engineering, and sustainability.

Research & Development

- Process Automation: Implement automation technologies in the 50-litre bioreactor to ensure consistency and scalability.
 - Refine the electrolysis process to reduce energy consumption and improve the purity of metal recovery.
 - Scale-up to 50-Litre Bioreactor: Design and test a 50-litre bioreactor based on the successful prototypes of smaller scales, integrating advanced process control systems.
- Environmental Impact Studies: Conduct life-cycle assessments and environmental impact studies to validate the green credentials of the technology.

Q3-Q4:

- Business Development: Strengthen partnerships in Hong Kong, Singapore, Philippines, and Malaysia.
- Team Hiring: Recruit a marketing and sales team to build brand and product awareness.
- Revenue: Initiate pre-sales activities and pilot project collaborations.

Product Development

- Key Milestone: Develop the electrolysis process and integrate it with the bioextraction process.
- Pilot 200-Litre Bioreactor: Design and commission a 200-litre bioreactor to demonstrate scalability and operational feasibility.
- Multi-Feedstock Processing: Begin testing the system with various types of e-waste and other metal-bearing waste streams.

- Partnership Development: Collaborate with academic and industrial partners to enhance technology development and gain feedback.

Year 3: 2026 - Pilot Testing and Validation

Q1-Q2:

- Revenue Projection: Aim to generate \$5 - 10 million by the end of Q2.
- Fundraising: Secure grants or additional investment to close Series A.
- Full-Scale Pilot Operation: Operate the 200-litre bioreactor continuously, refining process parameters and technology.

Q3-Q4:

- Team Expansion: Scale up operations, customer support, and sales teams.
- Business Development: Finalize commercial partnerships and supplier agreements.
- Market Trials: Initiate commercial trials with key industry partners to demonstrate the technology's effectiveness and to refine further based on user feedback.
- Regulatory Approvals: Start the process of obtaining necessary regulatory approvals for commercial deployment

Year 4: 2027 - Commercialization and Market Entry

Q1-Q2:

- Revenue: Aim to double the Annual Recurring Revenue (ARR) from pilot projects and early contracts.
- Business Development: Launch marketing campaigns and participate in industry events.
- Product Development
 - Launch of Commercial Bioreactors: Begin production and sale of commercial-scale bioreactors and associated technology.
 - Expansion of Slurry Electrolysis: Scale up the slurry electrolysis system to handle increased throughput from commercial operations.
- Global Market Penetration: Establish sales and service networks in key markets worldwide.

Q3-Q4:

- Team Expansion: Continue to grow the team to support operational and sales activities.
- Revenue: Intensify sales efforts and expand the customer base.
- Product Development: Implement ongoing enhancements to both bioextraction and electrolysis processes based on operational data.

- Sustainability Initiatives: Integrate renewable energy sources into the process chain to further reduce the carbon footprint.

Year 5: 2028 - Expansion and Growth

Q1-Q4:

- Revenue: Focus on growing revenue in both existing and new markets and exploring new applications of our technology.
- Team Expansion: Invest in acquiring talent, particularly in sales, marketing, and customer support.
- Fundraising: Explore a Series B funding round or other financing options to fuel expansion.
- Business Development:
 - Product and Research Development
 - Research and develop next generation bioextraction and electrolysis technologies, focusing on even lower energy requirements and higher efficiency.
 - Explore new applications and markets for the technology, including medical waste recycling and precious metal recovery from industrial waste.
- Strategic Partnerships and Acquisitions:
 - Deepen and broaden partnerships, continuously innovate product lines, and invest in customer success to ensure satisfaction and retention.
 - Look for strategic partners and acquisition targets to expand market reach and technology portfolio.

4.7. Distribution Channels

BioMetallica's distribution channels are strategically selected to efficiently reach our target customers within the precious metal recovery industry with a strong ESG focus and net zero roadmap. These channels are designed to ensure that the precious metals recovered through BioMetallica's innovative and sustainable processes are accessible to all our target customers. The effectiveness of these channels lies in their ability to align with the needs of the target market, optimize logistics, and maintain the quality and integrity of the BioMetallica's products and services. This section outlines brief descriptions of the primary distribution channels BioMetallica employs:

Direct Sales

- BioMetallica will engage in direct sales to large corporate customers, such as major precious metal recyclers, electronics manufacturers, medical device manufacturers and pharmaceutical companies.

This involves direct communication and negotiation, allowing for customized service offerings and bulk sales agreements.

- Direct sales enable BioMetallica to build strong relationships with key customers, providing personalized attention and tailored solutions. This approach is effective for engaging with large enterprises that value direct interaction and require specific metal purities or volumes.

Distribution Partners

- The company partners with specialized distributors and brokers that have established networks within the healthcare, recycling and metal trading sectors. These partners facilitate the distribution of BioMetallica's precious metals to smaller manufacturers and research institutions.
- Distribution partners extend BioMetallica's reach into markets and regions that might be challenging to serve directly. They leverage their logistics expertise and local market knowledge, ensuring efficient and timely delivery to customers who might otherwise be difficult to reach.

Strategic Alliances

- BioMetallica forms strategic alliances with companies and organizations in related fields, such as electronic waste recycling firms and research institutions. These alliances sometimes involve collaborative efforts in metal recovery and recycling initiatives. We currently have strategic alliance with iZ Precious and LEM industries in Italy.
- Strategic alliances not only broaden the distribution network but also enhance the value proposition of BioMetallica's offerings through collaboration and innovation. They are effective in creating new opportunities for the application of recovered precious metals and in establishing BioMetallica as a leader in sustainable recovery practices.

Industry Events and Trade Shows

- Participation in industry events, conferences, and trade shows related to cleantech, sustainable manufacturing, circular economy and sustainability serves as a channel for showcasing BioMetallica's products and technologies.
- These events provide a platform for direct engagement with potential customers, partners, and industry influencers. They are effective in generating leads, building brand awareness, and establishing BioMetallica's presence in the market.

BioMetallica's distribution channels are carefully chosen to support our business objectives, ensuring that our innovative and sustainable precious metal recovery solutions are accessible to the target markets. The effectiveness of these channels lies in their ability to facilitate direct engagement with customers, leverage online and technological tools for broader accessibility, and utilize partnerships to expand market reach and enhance service offerings.

4.8 Commercialization Plans

Upon successful completion of our Proof of Concept (POC) project, BioMetallica is positioned to transform the precious metal recycling sector with our pioneering, sustainable recovery process. Our commercialization strategy focuses on rapid scale-up, market penetration, and establishing leadership in environmentally friendly metal recovery solutions.

Business Model:

BioMetallica plans to implement a two-pronged commercialization approach, featuring Recycling-as-a-Service (RaaS) for corporate clients and technology licensing for extended reach.

- **RaaS:** Tailored to meet the needs of industries such as precious metal recycling, electronic manufacturing, automotive, and biomedical sectors, this model offers a sustainable, cost-effective alternative to conventional recycling methods. Targeting large firms committed to ESG principles and net-zero objectives, our services promise competitive gross margins of 30-50%, contingent upon the waste stream's metal content and composition. For every 1 kg of feedstock recycled, we can recover 27.8 g of precious metals of varying percentages depending on the composition of the feedstock.

- **Technology Licensing:** To broaden the impact of our technology, we will license our patented bioextraction processes to global recycling facilities. This strategy not only diversifies revenue but also promotes widespread adoption of eco-friendly recycling practices. A pivotal licensing condition is the allocation of all GHG emission reduction units generated by our technology at licensee facilities to BioMetallica.

5. Management

5.1 Management Team Biographies

For BioMetallica to succeed in the competitive field of sustainable precious metal recycling, it needs a diverse and skilled management team. Each member brings specific expertise, experience, and vision to navigate the challenges of starting and scaling an innovative business. This chapter provides details of BioMetallica's current and future team members, the roles and contributions they bring in providing BioMetallica with a competitive edge

CEO - Chief Executive Officer – Dr. Ing. Kwadwo “Joe” Konadu Ansah-Antwi

Background: Joe, a visionary leader with over 15 years in decarbonizing technologies and business



development, co-founded BioMetallica after leading Hessner Technologies.

There, he spearheaded the creation of an advanced energy storage system for low-speed electric vehicles in Africa and Southeast Asia. Motivated by a personal battle with asthma exacerbated by e-waste burning in Accra, Ghana, Joe is deeply committed to addressing the environmental and health impacts of electronic waste disposal. Joe holds a PhD in Electrical and

Computer Engineering and a Graduate Diploma in Management of Technology from the National University of Singapore. He is also a former research scientist at the University of Hong Kong.

- **Contribution:** Provides strategic direction, ensures alignment with the company's mission and values, and secures funding. Joe's experience in startup environments aids in navigating the challenges of scaling BioMetallica's innovative business model.

Chief Scientific Officer – Dr. Li Xirui

Background: Xirui brings over a decade of expertise in chemistry, biochemistry, and genetic engineering, marked by innovative contributions to metal recovery and remediation through the genetic modification of microorganisms. Xirui holds a PhD in Chemistry from the Nyanyang

Technological University, Singapore and a postdoctoral research fellow at the National University of Singapore.



- **Contribution:** Xirui leads the R&D efforts to develop and refine BioMetallica's biorecovery and purification technologies. Her expertise ensures that BioMetallica stays at the forefront of sustainable technology, maintaining a competitive edge through innovation.

Advisory Team

Mr. Rajeev Kumar: Rajeev is a distinguished Business Development Advisor at BioMetallica, bringing over four decades of expertise in the Chemical and Industrial Engineering sectors. A University of Delhi alumnus with a B.Sc and MBA, his career since 1979 has been marked by enhancing global production efficiencies and fostering international market expansions. At BioMetallica, Rajeev's unparalleled blend of technical knowledge and commercial insight has been instrumental in refining business strategies, driving profit growth, and aligning corporate objectives with global ambitions. His deep understanding of the international business environment makes him a pivotal asset in steering the company towards achieving its vision.

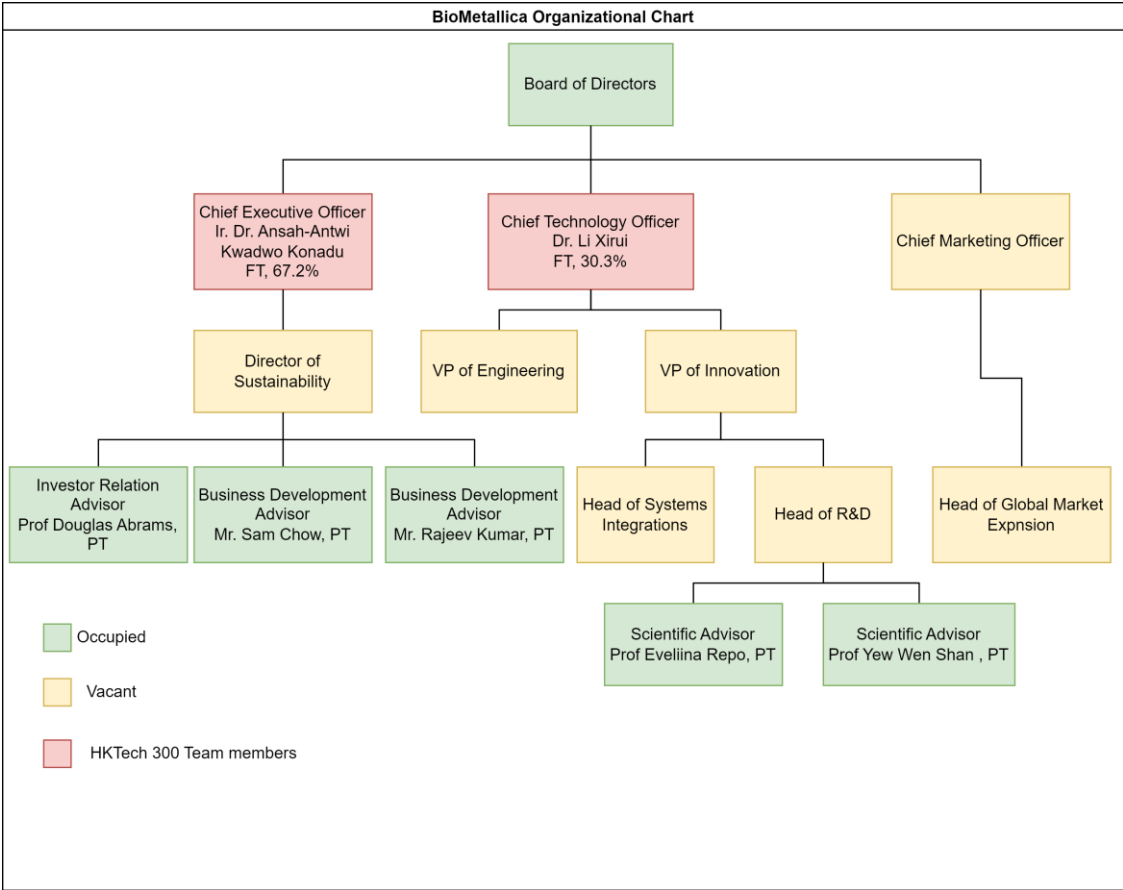


Mr. Sam Chow: Sam is a Professional Chartered Accountant (ACCA) is the founder of Star Metal Recall Sdn Bdn, Malaysia. Star Metall Recall specialises in developing and directing the implementation of a comprehensive scrap/waste management strategy named Total Scrap management for the aviation and oil & gas industries. Coming out from a Finance background, he has worked to build value-added relationships with the customer base and made a commitment to build trust and establish long-term relationships. In January 2010, Sam's team successfully introduced Total Scrap Management to our valued customers and the management system has been certified with ISO 9001:2008 in the same year. Our direct involvement in the scrap and waste industry for the past 22 years has given us the experience and knowledge to gain competitive advantage as professionals. This is our winning formula to meet our client objectives



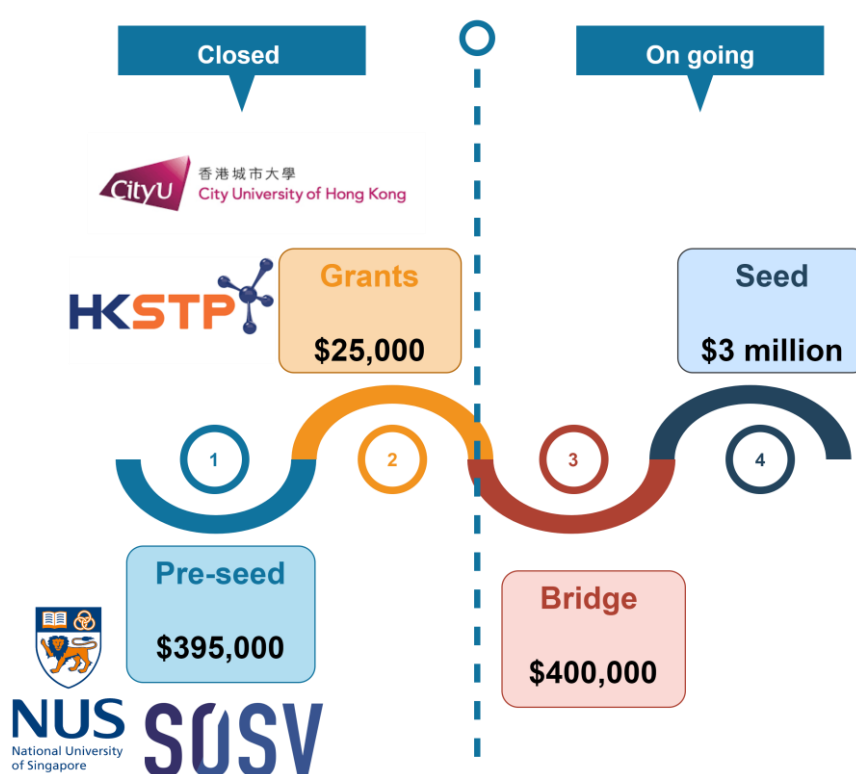
especially in high value alloy scrap revenue recovery as well as continuous reduction of waste disposal cost.

5.2 Organizational Chart



6. Financials

Since its inception, BioMetallica has successfully secured \$395,000 in funding from SOSV, a US-based venture capital firm, and NUS-Enterprise, the venture arm of the National University of Singapore. Additionally, the company has obtained \$25,000 in non-dilutive grants, further bolstering our financial resources for growth and development. The company is actively pursuing bridge financing of \$500,000, with aspirations to conclude this funding round by the end of Q1 2024. Following this, we plan to initiate our seed funding round in Q3/Q4 of 2024, aiming to raise approximately \$2-3 million through a priced round. This strategic financial planning is set to support our continued growth and innovation in sustainable metal recovery technologies.



BioMetallica's financial projections underscore our commitment to pioneering sustainable solutions in the precious metal recovery industry. Our business model, centered around innovative biotechnology, not only sets a new standard for environmental responsibility but also positions us for strong financial performance. Through strategic investments in research and development, efficient operational processes, and targeted market penetration, we forecast robust growth and profitability. Our financial strategy is designed to maximize returns for our stakeholders while minimizing environmental impact, reflecting our dual focus on fiscal responsibility and sustainability.

As we move forward, BioMetallica aims to capture a significant share of the precious metal recovery market, leveraging our competitive advantages in cost-effectiveness, sustainability, and technological innovation. Our financial forecasts are built on conservative estimates, ensuring realistic and attainable financial goals. These projections include detailed cash flow forecasts, a comprehensive break-even analysis, and key financial ratios that demonstrate our commitment to operational excellence and financial health. With a clear path to profitability and a strong foundation in sustainable technology, BioMetallica is poised to lead the industry towards a greener future, creating value for our investors, customers, and the planet.

Below is a snapshot of some key financial metrics and indicators:

BioMetallica HK Ltd			Subsidiary		5			
Financial Statements- Annual			USD:HKD		7.82			
			2024	2025	2026	2027	2028	2029
			Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Revenue	USD	0	2,674,029	39,448,647	81,299,713	137,925,518	264,080,129	
Recycling Service	HKD	0	564,792	7,335,879	14,642,062	25,270,901	46,528,580	
Precious metal trading	HKD	0	2,109,237	29,532,168	58,915,851	104,912,817	196,906,750	
Carbon credits	HKD	0	0	1,032,240	3,096,720	3,096,720	8,257,920	
Technology Licensing	HKD	0	0	1,548,360	4,645,080	4,645,080	12,386,880	
COGS	HKD	0	1,445,253	21,071,081	38,252,073	67,462,590	130,983,900	
Recycling Service	HKD	0	817,278	10,056,795	17,800,791	29,286,863	54,632,447	
Precious metal trading	HKD	0	627,975	11,014,285	20,451,282	38,175,727	48,818,065	
Other COGS	HKD	0	0	0	0	0	0	
Gross Profit	HKD	0	1,228,776	18,377,566	43,047,639	70,462,928	133,096,229	
Gross margin	%	-	46.0%	46.6%	52.9%	51.1%	50.4%	
SG&A	HKD	269,978	1,152,249	1,795,762	1,937,577	2,090,476	2,258,666	
Payroll	HKD	269,978	830,690	1,488,529	1,627,529	1,780,429	1,948,619	
Marketing & Growth	HKD	0	1,877	1,877	1,877	1,877	1,877	
Advisory & Professional Services	HKD	0	15,640	0	0	0	0	
Rent	HKD	0	281,520	281,520	281,520	281,520	281,520	
Tech Support & Services	HKD	0	1,877	3,191	6,006	6,006	6,006	
Insurance	HKD	0	9,384	9,384	9,384	9,384	9,384	
Utilities	HKD	0	11,261	11,261	11,261	11,261	11,261	
Other Expenses	HKD	0	0	0	0	0	0	
EBITDA	HKD	(269,978)	76,527	16,581,805	41,110,063	68,372,452	130,837,563	
EBITDA margin	%	-	2.9%	42.0%	50.6%	49.6%	49.5%	
D&A	HKD	2,520	8,254	236,469	695,289	1,610,668	3,407,703	
EBIT	HKD	(272,497)	68,273	16,345,335	40,414,774	66,761,784	127,429,860	
EBIT margin	%	-	2.6%	41.4%	49.7%	48.4%	48.3%	
Interest expense	HKD	0	0	0	0	0	0	
Income taxes	HKD	0	243,094	3,586,241	8,666,792	15,841,073	29,429,594	
Net income	HKD	(272,497)	(174,821)	12,759,095	31,747,982	50,920,711	98,000,265	
Net margin	%	-	0.0%	32.3%	39.1%	36.9%	37.1%	

BALANCE SHEET

Excess cash	HKD	(279,362)	2,660,176	13,833,836	58,310,835	103,629,672	268,812,556
Inventories	HKD	0	0	0	0	0	0
Prepaid expenses	HKD	0	0	0	0	0	0
Current Assets	HKD	(279,362)	2,660,176	13,833,836	58,310,835	103,629,672	268,812,556
PP&E (net)	HKD	6,864	20,506	1,605,940	4,516,923	10,118,797	21,136,179
Intangible assets	HKD	0	0	0	0	0	0
Investments	HKD	0	0	0	0	0	0
Other assets (net)	HKD	0	0	0	0	0	0
Non-Current Assets	HKD	6,864	20,506	1,605,940	4,516,923	10,118,797	21,136,179
Total Assets	HKD	(272,497)	2,680,682	15,439,777	62,827,758	113,748,469	289,948,734
Financial obligations (Debt)	HKD	0	0	0	0	0	0
Total Liabilities	HKD	0	0	0	0	0	0
Common equity	HKD	0	3,128,000	3,128,000	18,768,000	18,768,000	96,968,000
Net earnings	HKD	(26,864)	58,704	1,060,136	1,471,258	1,208,002	3,181,338
Retained earnings	HKD	(245,633)	(506,022)	11,251,641	42,588,500	93,772,467	189,799,396
Total Equity	HKD	(272,497)	2,680,682	15,439,777	62,827,758	113,748,469	289,948,734
Total Liabilities + Equity	HKD	(272,497)	2,680,682	15,439,777	62,827,758	113,748,469	289,948,734
		0.0	0.0	0.0	0.0	0.0	0.0

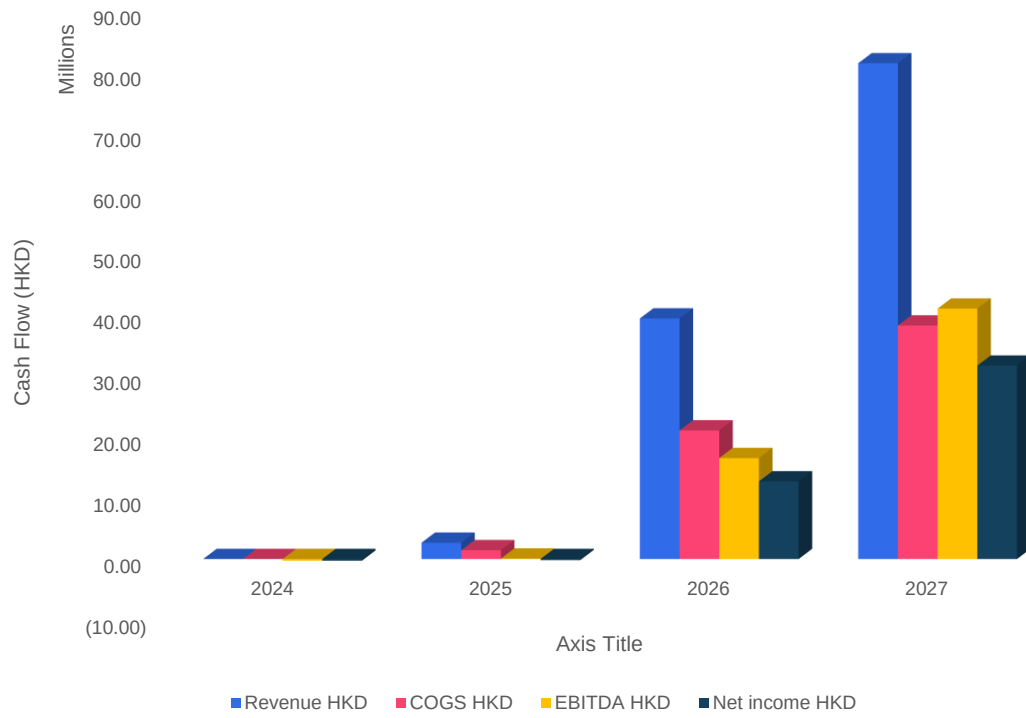
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CASH FLOW

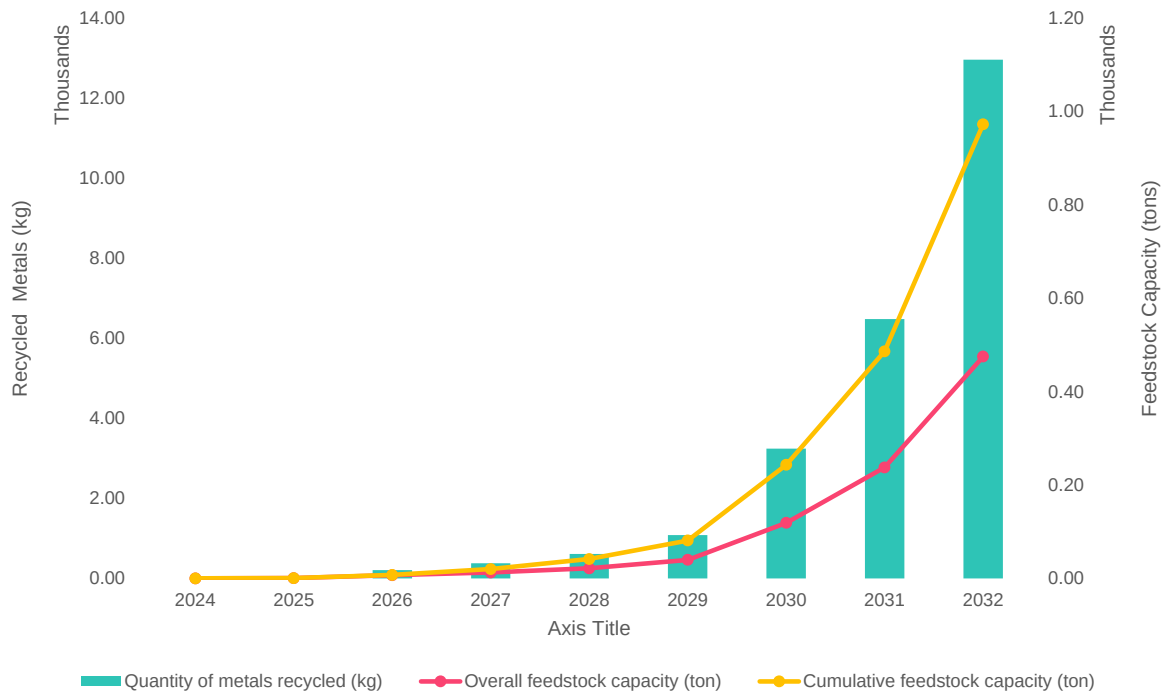
Initial cash balance	0	HKD	0	(279,362)	2,660,176	13,833,836	58,310,835	103,629,672
Funding Round								
Cash infusions		HKD	0	3,128,000	0	15,640,000	0	78,200,000
Operating Change in Cash		HKD	(279,362)	(188,462)	11,173,661	28,836,999	45,318,837	86,982,884
Net income		HKD	(272,497)	(174,821)	12,759,095	31,747,982	50,920,711	98,000,265
(+) D&A		HKD	2,520	8,254	236,469	695,289	1,610,668	3,407,703
(-) CAPEX		HKD	9,384	21,896	1,821,904	3,606,271	7,212,542	14,425,085
(-) Dividends Paid to Investors		HKD	0	0	0	0	0	0
Ending cash balance			(279,362)	2,660,176	13,833,836	58,310,835	103,629,672	268,812,556

Graphical illustrations of key financial indicators of BioMetallica's business projected over the next 5 years are shown below:

Financial Projection



Processing Capacity Projection



7. Offering/Funding Request

7.1 Offer

BioMetallica is seeking an investment of \$2-3 million to accelerate the development and deployment of our groundbreaking biotechnology for precious metal recovery. We are offering a 10% equity stake in our company, valuing BioMetallica at \$20 million. We propose a flexible deal structure that could include a combination of equity, convertible notes, or a straight equity investment, with detailed terms to be negotiated based on investor preference. We anticipate a strong return on investment, driven by our unique positioning in the market, with projected profitability within three years of full-scale operations. Our offer includes comprehensive support and transparent communication with our investors, ensuring a clear understanding of our business operations and growth strategy.

7.2 Capital Requirements

The \$3 million funding will be utilized to finalize R&D efforts, scale up our production capabilities, and expand our market reach. Specifically, funds will go towards enhancing our laboratory infrastructure, purchasing essential equipment, and hiring key personnel to support our growth. This investment will enable BioMetallica to meet the growing demand for sustainable precious metal recovery, offering mutual benefits through significant growth potential and profitability. Our partners will benefit not only from financial returns but also by being part of a solution that addresses critical environmental challenges.

7.3 Risk/Opportunity

BioMetallica acknowledges the inherent risks in pioneering new technologies in the recycling sector, including market adoption rates and operational scalability. However, we've mitigated these risks through extensive market research, pilot testing with potential customers, and securing strategic partnerships within the industry. The opportunity for BioMetallica far outweighs these risks, given the urgent demand for sustainable metal recovery solutions and the growing global focus on environmental sustainability. Our innovative approach positions us to capture a significant market share and establish BioMetallica as a leader in the industry.

7.4 Valuation of Business

The valuation of BioMetallica at \$20 million for our upcoming seed priced round is based on a comprehensive analysis, including our proprietary technology's potential, projected cash flows, market size, and comparative valuations of similar companies in the sector. This valuation reflects our current market position, intellectual property, and the future growth trajectory enabled by our innovative

recovery process. We are committed to driving value for our investors by executing our strategic plan and achieving our financial targets.

7.5 Exit Strategy

BioMetallica is committed to providing a lucrative exit strategy for our investors. Options include a buy-back agreement, where the company would repurchase shares from investors at a predetermined rate of return or facilitating a sale to a strategic buyer or another investor. Additionally, we aim to explore the possibility of an initial public offering (IPO) in NASDAQ as the business scales, offering investors a pathway to liquidity through the public markets at a proposed valuation of \$1.5 billion. Our exit strategy is designed to ensure that investors have multiple avenues to realize their investment's value while supporting BioMetallica's long-term vision and sustainability goals.